

FCA Mid-Semester Hackathon: Catch the Rocket

Complete Engineering, Mathematical Architecture & Hardware Verification Report

Autonomous Ground-Station Firmware on a Physical ESP32 (Levels 0–5)

Team ByteForce

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Abstract

This report documents how **Team ByteForce** developed, verified and hardened an autonomous ground station on an ESP32 microcontroller (Xtensa LX6, 240 MHz) for the FCA “Catch the Rocket” hackathon. The tracker models a 100-second simulated flight delivered as exactly 100,000 altitude readings: reading index i corresponds to simulated time $t = i/1000$ s, i.e. the trajectory is sampled at 1 kHz (one reading per simulated millisecond). Separately, per the hackathon problem specification, the readings are delivered to the ground station at a burst rate of 20,000 readings per second (one reading every 50 μ s). These are two different rates: the first describes the simulated flight, the second sets the processor’s per-reading deadline of $240 \text{ MHz} \times 50 \mu\text{s} = 12,000$ CPU cycles.

The system keeps all persistent reading-derived state in one C struct of at most 4096 bytes (no flash, SPIFFS, NVS or wireless), processes every reading within the 12,000-cycle deadline, hard-codes no flight parameters, and answers all eight serial commands (STATS, G, APOGEE, DEPLOY, DESCENT, LAND, GLITCHES, ALT t) with measured cycle latencies. We completed Levels 0–5. On the Official Tracker and Practice Tracker B, running on physical ESP32 hardware, the final Level 5 firmware reached a worst-case per-reading cost of **6502 cycles (54.2% of the budget)** and a state size of **2336 bytes (57.0% of 4096; 1760 bytes free)**. All reported flight values are within the official judging tolerances.

Submission Summary

- **Archive:** Team-ByteForce.zip with 11 sketches (LEVEL_0_FINAL.ino ... LEVEL_5_OFFICIAL.ino / PRACTICE.ino) and the serial-monitor captures (outputs.docx).
- **Problem Specification:** fca_hackathon_catch_the_rocket.pdf.
- **Hardware Bench:** Espressif ESP32 Dev Module (240 MHz, single-precision hardware FPU, 520 KB SRAM), serial monitor at 115,200 baud on COM3.
- **Official Flight (measured):** $g = 9.810 \text{ m/s}^2$ (truth 9.810); apogee 8.002 s / 626.08 m; deployment 12.000 s / 547.68 m; descent -4.500 m/s ; landing 133.706 s (truth 133.707 s); glitches 739.
- **Practice Flight B (measured):** $g = 3.711 \text{ m/s}^2$ (truth 3.710); apogee 11.321 s / 1737.75 m; deployment 23.999 s / 1439.54 m; descent -8.500 m/s ; landing 193.351 s (truth 193.355 s); glitches 727.

1 Project Overview, Rules and Requirements

The objective is a real-time autonomous ground station on an ESP32 that tracks a rocket flight. The tracker supplies exactly 100,000 successive altitude readings through calls to `nextAltitude()`, one per simulated millisecond ($t = i/1000$ s).

1.1 Physical Guarantees (G1–G4)

- **G1 (readings):** every reading is a whole number of centimetres between 0 m and 3000 m.
- **G2 (flight mechanics):** (a) coast phase ($t < t_D$): unpowered ballistic flight $h(t) = h_0 + v_0 t - \frac{1}{2}gt^2$ with $g \in [2.0, 30.0]$ m/s²; (b) parachute deployment at t_D between 8 s and 45 s, apogee at least 1.0 s before deployment; (c) terminal descent ($t \geq t_D$) at constant $v_D \in [-15, -2]$ m/s, and at deployment the rocket falls at least 10 m/s faster than under parachute; (d) the rocket is still at least 20 m up at $t = 100$ s.
- **G3 (sensor noise):** clean telemetry is within ± 2.00 m (± 200 cm) of the true trajectory.
- **G4 (glitches):** active when `TRACKER_GLITCHES` is 1, for $t \geq 1.0$ s; at most 1% of readings (at most 1000), errors of 50 – 300 m, spaced at least 10 clean readings apart.

1.2 Embedded-Architecture Constraints

- **Timing deadline (12,000 cycles per reading):** Readings are delivered at 20,000 per second (a 50 μ s arrival interval). At 240 MHz this gives a per-reading deadline of $240 \times 50 = 12,000$ CPU cycles; processing must finish within it to avoid losing sync. This deadline is independent of the 1 kHz simulated-flight sampling.
- **Single-struct memory limit (4096 bytes):** All state remembered across readings must live in one C struct of at most 4096 bytes. Storing the raw flight as floats would need 400 KB (prohibited). No flash, EEPROM, SPIFFS, NVS or wireless.
- **No hard-coded constants:** The code must not contain the tracker secrets (H_0, V_0, G, T_D, V_D) and must infer the trajectory from the stream.
- **Serial protocol:** After ingestion the firmware answers `STATS`, `G`, `APOGEE`, `DEPLOY`, `DESCENT`, `LAND`, `GLITCHES` and `ALT t` with cycle latencies; invalid queries print `ERR` without crashing.

1.3 Official Judging Tolerances

Table 1: Judging Limits and Tolerances

Command / Metric	Level 1	Level 2	Levels 3–5 (final)
DESCENT	± 0.01 m/s	± 0.01 m/s	± 0.01 m/s
LAND	± 0.50 s	± 0.50 s	± 0.50 s
G	–	± 0.05 m/s ²	± 0.02 m/s ²
APOGEE time / altitude	–	± 0.25 s / ± 10 m	± 0.02 s / ± 0.25 m
DEPLOY time / altitude	–	–	± 0.02 s / ± 0.25 m
ALT t	–	–	± 0.25 m
GLITCHES	–	–	exact match

2 Mission Control Workflow and Operating Protocols

Development followed a six-step closed loop for every level, with verification gates.

Table 2: Closed-Loop Development Cycle

Step	Activity	Verification Criterion
1	Mathematical planning	Algebraic derivation, coordinate-centering argument, overflow bounds
2	Drafting firmware	Non-blocking C/C++ inside the <code>GroundState</code> struct
3	Skeptic code audit	Numerical stability, integer limits, no dynamic memory
4	Hardware bench flash	Flash the ESP32, capture serial output (115,200 baud)
5	Verification against truth	Compare outputs with tracker truth and the judging limits
6	Documentation and freeze	Archive logs and cycle counts, commit the verified sketch

A build that exceeds 12,000 cycles or misses a limit is rejected and the loop restarts at step 1; a build that passes on both the Official Tracker and Practice Tracker B is documented, frozen and promoted.

Operating Protocols: Single-struct rule (any `GroundState` change needs unanimous approval); math first, code second; immutable logs (runs are appended with level and tracker tags); defense readiness (everyone can explain every derivation).

3 Level 0: Bring-Up, Skeleton and Cycle Harness

Level 0 proved that the ESP32 can take 100,000 consecutive `nextAltitude()` calls while measuring per-reading cost. The cycle counter is read around the processing of each reading; the tracker call itself is outside the measured section:

```
float alt = nextAltitude();           // tracker call: not measured
uint32_t tStart = ESP.getCycleCount();
/* firmware state processing */
uint32_t elapsed = ESP.getCycleCount() - tStart;
if (elapsed > state.worstCycles) state.worstCycles = elapsed;
```

A non-blocking serial parser with a 32-byte circular buffer returns `TODO` plus the command's cycle count; `STATS` prints memory use and the worst-case cycles.

3.1 Theoretical Baseline (Truth by Hand)

The Official Tracker secrets are $H_0 = 31200$ cm, $V_0 = 7850$ cm/s, $G = 981$ cm/s², $T_D = 12000$ ms, $V_D = -450$ cm/s. Then:

$$t_{\text{apogee}} = V_0/G = \mathbf{8.002 \text{ s}}, \quad H_{\text{apogee}} = H_0 + V_0^2/(2G) = \mathbf{626.08 \text{ m}},$$

$$t_D = \mathbf{12.000 \text{ s}} \text{ with } H_D = \mathbf{547.68 \text{ m}}, \quad v_D = \mathbf{-4.500 \text{ m/s}}, \quad t_{\text{land}} = \mathbf{133.707 \text{ s}}.$$

(theoretical reference truth).

For Practice Tracker B ($H_0 = 150000$, $V_0 = 4200$, $G = 371$, $T_D = 24000$, $V_D = -850$) the same formulas give $g = \mathbf{3.710 \text{ m/s}^2}$, apogee $\mathbf{11.321 \text{ s}/1737.74 \text{ m}}$, deployment $\mathbf{24.000 \text{ s}/1439.52 \text{ m}}$, descent $\mathbf{-8.500 \text{ m/s}}$, landing $\mathbf{193.355 \text{ s}}$. These “truth” values are used only to judge results; the firmware never contains them.

Table 3: Level 0 Hardware Measurements (ESP32, 240 MHz)

Metric	Official	Practice B	Limit	Status
Readings processed	100,000 / 100,000	100,000 / 100,000	100,000	PASS
State memory	44 B	44 B	≤ 4096 B	PASS
Worst reading	903 cycles	903 cycles	$\leq 12,000$	PASS
Headroom	11,097 (92.5%)	11,097 (92.5%)	–	–
Serial parser	functional (TODO / ERR)	functional	–	PASS

```
Output Serial Monitor X
Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')
mode:010, clock_div:1
load:0x3fff0030,len:4876
ho 0 tail 12 room 4
load:0x40078000,len:16532
load:0x40080400,len:3500
entry 0x400805b4

=====
CATCH THE ROCKET - LEVEL 0
=====
Receiving 100000 readings...

Receiving complete.
Readings received: 100000
READY
TODO
command cycles: 579
TODO
command cycles: 14
TODO
command cycles: 14
TODO
command cycles: 14
TODO
command cycles: 14
TODO
command cycles: 9
TODO
command cycles: 14
bytes used: 44 of 4096
worst reading: 903 cycles
command cycles: 12

=====
```

Figure 1: Level 0 serial monitor: 100,000 readings received, 44 bytes of state, worst reading 903 cycles.

4 Level 1: The Descent (Linear Regression and Touchdown)

4.1 Mathematical Formulation

By G2 the parachute has opened by $t = 45$ s ($i = 45,000$), so for $t \geq 45$ s the altitude is a straight line $h(t) = h_D + v_D(t - t_D)$. Fitting it with raw millisecond indices $i \in [45000, 99999]$ makes $\sum i^2$ about 3.0×10^{14} , which is large enough to make double-precision round-off visible. Chaithra Gana G therefore introduced a centred time coordinate:

$$k = i - 72500 \text{ (ms)}, \quad k \in [-27500, +27500).$$

Because k is symmetric about the middle of the window, $\sum k = 0$ and the normal equations decouple. With $y = A + Bk$, $N = 55,000$ and five running sums ($N, \sum k, \sum y, \sum k^2, \sum ky$):

$$B = \frac{\sum ky}{\sum k^2} \text{ (cm/ms)}, \quad A = \frac{\sum y}{N} \text{ (cm)}.$$

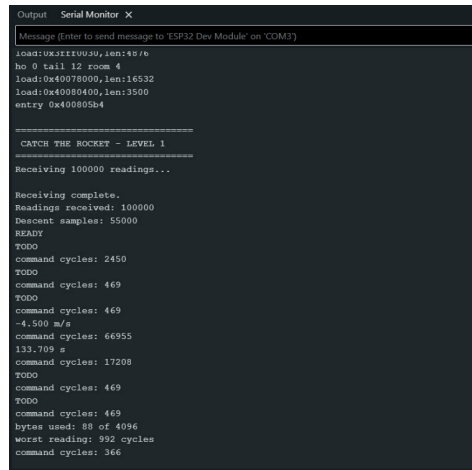
Since $1 \text{ cm/ms} = 10 \text{ m/s}$, the terminal velocity and touchdown time ($y = 0$) are:

$$v_D = 10B \text{ (m/s)}, \quad k_{\text{land}} = -A/B, \quad t_{\text{land}} = \frac{72500 + k_{\text{land}}}{1000} \text{ (s)}.$$

Table 4: Level 1 Hardware Results vs Truth

Metric	Official Measured	Official Truth	Practice B Measured	Practice B Truth
DESCENT	-4.500 m/s	-4.500 m/s	-8.500 m/s	-8.500 m/s
LAND	133.709 s	133.707 s	193.357 s	193.355 s
State memory	88 B	≤ 4096 B	88 B	≤ 4096 B
Worst reading	992 cycles	$\leq 12,000$	992 cycles	$\leq 12,000$

Both flights matched truth with zero error on velocity and 2 ms error on touchdown time, using 8.3% of the cycle budget.



```

Output Serial Monitor X
Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')
load:0x3f100000,len:4096
ho 0 tail 12 room 4
load:0x40078000,len:16532
load:0x40080400,len:3500
entry 0x400805b4

=====
CATCH THE ROCKET - LEVEL 1
=====
Receiving 100000 readings...

Receiving complete.
Readings received: 100000
Descent samples: 55000
READY
TODO
command cycles: 2450
TODO
command cycles: 469
TODO
command cycles: 469
-4.500 m/s
command cycles: 66955
133.709 s
command cycles: 17208
TODO
command cycles: 469
TODO
command cycles: 469
bytes used: 88 of 4096
worst reading: 992 cycles
command cycles: 366

```

Figure 2: Level 1, Official Tracker: DESCENT -4.500 m/s, LAND 133.709 s, 88 bytes, worst reading 992 cycles.

```
Output  Serial Monitor  X
Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')
load:0x3f10030, len:48 /b
ho 0 tail 12 room 4
load:0x40078000, len:16532
load:0x40080400, len:3500
entry 0x400805b4

=====
CATCH THE ROCKET - LEVEL 1
=====
Receiving 100000 readings...

Receiving complete.
Readings received: 100000
Descent samples: 55000
READY
TODO
command cycles: 2168
TODO
command cycles: 469
TODO
command cycles: 469
-8.500 m/s
command cycles: 66916
193.357 s
command cycles: 17208
TODO
command cycles: 469
TODO
command cycles: 469
bytes used: 88 of 4096
worst reading: 992 cycles
command cycles: 366
```

Figure 3: Level 1, Practice Tracker B: DESCENT -8.500 m/s, LAND 193.357 s, 88 bytes, worst reading 992 cycles.

5 Level 2: The Coast (Ballistic Flight, Gravity and Apogee)

5.1 Mathematical Formulation

By G2 the parachute never opens before $t = 8$ s ($i = 8000$), so the first 8000 readings are pure ballistic coast: $y(t) = a + bt + ct^2$. With the centred coordinate:

$$q = 2i - 7999, \quad i \in [0, 7999] \implies q \in \{-7999, -7997, \dots, +7997, +7999\},$$

all odd power sums vanish: $\sum q = 0$ and $\sum q^3 = 0$. This diagonalises the 3×3 normal equations (with $y = A + Bq + Cq^2$):

$$B = \frac{\sum qy}{\sum q^2}, \quad C = \frac{N \sum q^2 y - (\sum q^2)(\sum y)}{N \sum q^4 - (\sum q^2)^2}, \quad A = \frac{\sum y - C \sum q^2}{N}.$$

Because $q = 2000t - 7999$ (t in seconds), $d^2y/dt^2 = C(dq/dt)^2 = 4 \times 10^6 C$ cm/s², so:

$$g = -80000C \text{ (m/s}^2\text{)}, \quad q_{\text{apogee}} = -\frac{B}{2C}, \quad t_{\text{apogee}} = \frac{7999 + q_{\text{apogee}}}{2000} \text{ (s)},$$

and the apogee altitude is $y(q_{\text{apogee}})$ converted to metres.

Skeptic Review (Gana): For $N = 8000$ the sums are $\sum q^2 = 1.707 \times 10^{11}$ and $\sum q^4 = 6.55 \times 10^{18}$ (below the int64 limit of 9.22×10^{18}). The determinant $N \cdot \sum q^4 - (\sum q^2)^2$ subtracts two numbers of order 5×10^{22} and 3×10^{22} and equals about 2.3×10^{22} . A 32-bit float has a 24-bit mantissa (about 7 significant digits), so this subtraction loses all significant digits. Gana therefore mandated 64-bit integer accumulators (`int64_t`), exact up to 9.22×10^{18} , and double-precision arithmetic for the division.

Table 5: Level 2 Hardware Results vs Truth

Metric	Official Measured	Official Truth	Practice B Measured	Practice B Truth
Gravity	9.814 m/s ²	9.810 m/s ²	3.709 m/s ²	3.710 m/s ²
Apogee time	8.001 s	8.002 s	11.325 s	11.321 s
Apogee altitude	626.07 m	626.08 m	1737.81 m	1737.74 m
State memory	136 B	≤ 4096 B	136 B	≤ 4096 B
Worst reading	2404 cycles	$\leq 12,000$	3343 cycles	$\leq 12,000$

```

Output  Serial Monitor X
[Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')]

no u tail iz room 4
load:0x40078000,len:16532
load:0x40080400,len:3500
entry 0x400805b4

=====
CATCH THE ROCKET - LEVEL 2
=====
Receiving 100000 readings...

Receiving complete.
Readings received: 100000
Coast samples: 8000
Descent samples: 55000
READY
9.814 m/s^2
command cycles: 66972
8.001 s 626.07 m
command cycles: 24065
TODO
command cycles: 582
-4.500 m/s
command cycles: 14020
133.709 s
command cycles: 16064
TODO
command cycles: 10
TODO
command cycles: 16
bytes used: 136 of 4096
worst reading: 2404 cycles
command cycles: 289

```

Figure 4: Level 2, Official Tracker: gravity 9.814 m/s^2 , apogee $8.001 \text{ s} / 626.07 \text{ m}$.

```

Output  Serial Monitor X
[Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')]

load:0x40078000,len:16532
load:0x40080400,len:3500
entry 0x400805b4

=====
CATCH THE ROCKET - LEVEL 2
PRACTICE TRACKER B
=====
Receiving 100000 readings...

Receiving complete.
Readings received: 100000
Coast samples: 8000
Descent samples: 55000
READY
3.709 m/s^2
command cycles: 65953
11.325 s 1737.81 m
command cycles: 26610
TODO
command cycles: 584
-8.500 m/s
command cycles: 14883
193.357 s
command cycles: 15784
TODO
command cycles: 10
TODO
command cycles: 16
bytes used: 136 of 4096
worst reading: 3343 cycles
command cycles: 5

upload sketch

```

Figure 5: Level 2, Practice Tracker B: gravity 3.709 m/s^2 , apogee $11.325 \text{ s} / 1737.81 \text{ m}$.

6 Level 3: Block Compression, Deployment and Altitude Queries

6.1 Block-Based Moment Compression (2000 Bytes)

Level 3 had to find the deployment point and answer ALT t for any $t \in [0, 100]$ s. Because 100,000 readings cannot fit in 4096 bytes, the stream was compressed into 100 blocks of 1000 readings (1.0 s per block). For block $b \in [0, 99]$ with local coordinate $w = 2j - 999$:

$$s_0[b] = \sum_{j=0}^{999} h_j, \quad s_1[b] = \sum w \cdot h_j, \quad s_2[b] = \sum w^2 \cdot h_j.$$

s_0 is stored as `int32_t` (400 B) and s_1, s_2 as `int64_t` (800 B each): exactly 2000 bytes. **This block representation was an intermediate Level 3 design; the final Level 4–5 firmware does not store blocks (see Section 7 and Section 9).**

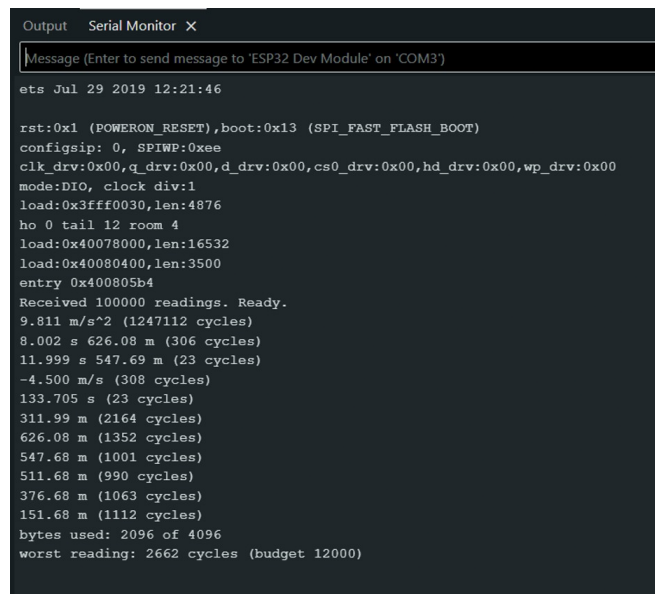
6.2 Curve Intersection and Root Selection

A parabola $P(t)$ is fitted to early blocks and a line $L(t)$ to late blocks, and $P(t) - L(t) = 0$ is solved. The quadratic has two roots. The earlier root is an unphysical artefact where the backward-extrapolated descent line crosses the ascending trajectory; the later root is the true parachute deployment (by G2 apogee precedes deployment, so the valid root is the larger one).

This level was run with `TRACKER_GLITCHES 0` (clean data).

Table 6: Level 3 Hardware Query Results (Clean Data)

Query	Official Output	Official Truth	Practice B Output	Practice B Truth
G	9.811 m/s ²	9.810	3.711 m/s ²	3.710
APOGEE	8.002 s, 626.08 m	8.002 s, 626.08 m	11.321 s, 1737.75 m	11.321 s, 1737.74 m
DEPLOY	11.999 s, 547.69 m	12.000 s, 547.68 m	23.999 s, 1439.54 m	24.000 s, 1439.52 m
DESCENT	-4.500 m/s	-4.500	-8.500 m/s	-8.500
LAND	133.705 s	133.707 s	193.351 s	193.355 s
ALT 0 s	311.99 m	312.00 m	1499.95 m	1500.00 m
ALT 12 s	547.68 m	547.68 m	1736.90 m	1736.88 m
ALT 50 s	376.68 m	376.68 m	1218.52 m	1218.52 m
ALT 100 s	151.68 m	151.68 m	793.51 m	793.52 m
State memory / worst reading		2096 B / 2662 cycles		2096 B / 2662 cycles



```

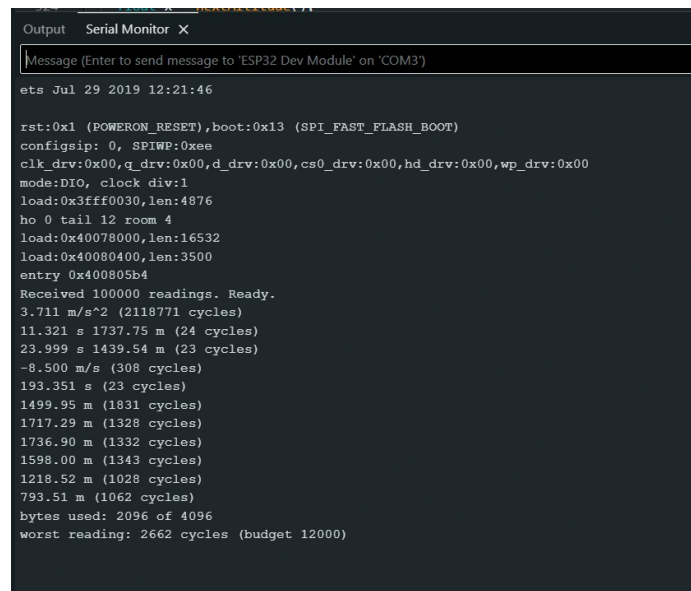
Output  Serial Monitor X
Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')

ets Jul 29 2019 12:21:46

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4876
ho 0 tail 12 room 4
load:0x40078000,len:16532
load:0x40080400,len:3500
entry 0x400805b4
Received 100000 readings. Ready.
9.811 m/s^2 (1247112 cycles)
8.002 s 626.08 m (306 cycles)
11.999 s 547.69 m (23 cycles)
-4.500 m/s (308 cycles)
133.705 s (23 cycles)
311.99 m (2164 cycles)
626.08 m (1352 cycles)
547.68 m (1001 cycles)
511.68 m (990 cycles)
376.68 m (1063 cycles)
151.68 m (1112 cycles)
bytes used: 2096 of 4096
worst reading: 2662 cycles (budget 12000)

```

Figure 6: Level 3, Official Tracker: deployment 11.999 s / 547.69 m and query outputs.



```

Output  Serial Monitor X
Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')

ets Jul 29 2019 12:21:46

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4876
ho 0 tail 12 room 4
load:0x40078000,len:16532
load:0x40080400,len:3500
entry 0x400805b4
Received 100000 readings. Ready.
3.711 m/s^2 (2118771 cycles)
11.321 s 1737.75 m (24 cycles)
23.999 s 1439.54 m (23 cycles)
-8.500 m/s (308 cycles)
193.351 s (23 cycles)
1499.95 m (1831 cycles)
1717.29 m (1328 cycles)
1736.90 m (1332 cycles)
1598.00 m (1343 cycles)
1218.52 m (1028 cycles)
793.51 m (1062 cycles)
bytes used: 2096 of 4096
worst reading: 2662 cycles (budget 12000)

```

Figure 7: Level 3, Practice Tracker B: deployment 23.999 s / 1439.54 m and query outputs.

7 Level 4: Glitch Rejection and Real-Time Ring Buffering

7.1 Causal 5-Point Median Filter and Quantization

In Level 4 glitches were switched on (`TRACKER_GLITCHES 1`): 50–300 m spikes (5000 to 30000 cm), at least 10 clean readings apart, on top of normal noise bounded by ± 2.00 m. Because future readings are not available while streaming, the filter is causal: a 5-sample window $r = [r_{i-4}, \dots, r_i]$ holds the current and four previous raw readings.

Altitudes are quantized to 5 cm units, $v = (h_{\text{cm}} + 2)/5$, so a `uint16_t` covers up to $65,535 \times 5 \text{ cm} = 3276 \text{ m}$ (the firmware clamps at 60,000 units = 3000 m, matching G1). A compact 5-value insertion sort computes $\text{med} = \text{median5}(r_{i-4}, \dots, r_i)$. If $|r_i - \text{med}| \times 5 \text{ cm} > 2000 \text{ cm}$ (20 m), the reading is flagged as a glitch, `glitchCount` is incremented and r_i is replaced by med . The glitch test is active from reading 1000 onward (glitches start at $t \geq 1.0 \text{ s}$).

Where precision is lost: The median-filtered value, in 5 cm units, is what feeds the ring buffer and the regression accumulators (the code uses $y = 5 \times v$ centimetres). The only precision lost in the data path is therefore the rounding to 5 cm, at most $\pm 2.5 \text{ cm}$ per stored reading, small compared with the $\pm 2 \text{ m}$ sensor noise that the regressions average out. Fit coefficients are kept in double precision and the sums in `int64`.

Skeptic safeguard (glitch vs deployment): An isolated 300 m spike cannot be mistaken for the parachute opening because (1) the median filter removes spikes before ring-buffer insertion and deployment testing, and (2) deployment is a persistent deviation lasting for the rest of the flight, whereas glitches are single-reading spikes at least 10 readings apart.

7.2 1024-Element Ring Buffer and Trajectory-Divergence Trigger

Cleaned readings enter a 1024-element circular buffer (`uint16_t ring[1024]`, 2048 bytes). Once the first 8-second fit is ready, and on every 16th reading, the firmware compares the actual altitude change over the last 1024 readings with the change predicted by the ballistic fit:

$$\Delta_{\text{actual}} = y_i - y_{i-1024}, \quad \Delta_{\text{pred}} = B(x_i - x_{i-1024}) + C(x_i^2 - x_{i-1024}^2),$$

triggering when $\Delta_{\text{actual}} - \Delta_{\text{pred}} > 500 \text{ cm}$.

The coast accumulators are fed with a 1024-reading delay (reading $i - 1024$ is added when reading i arrives), so by the time deployment is detected no transitional reading has entered the coast fit. After detection the transition window is frozen and the two curve fits are kept separate.

Why 500 cm works (derived, not measured): The difference of two noisy readings can differ from the truth by at most $2 \times 200 = 400 \text{ cm}$, below the 500 cm threshold. In the tracker model the velocity changes instantly at t_D , so with a velocity step $\Delta v \geq 10 \text{ m/s}$ the actual altitude lags the ballistic prediction by:

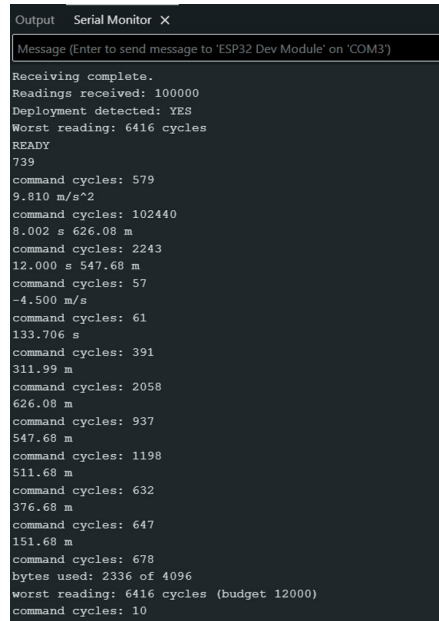
$$\Delta v \cdot \tau + \frac{1}{2} g \tau^2 \geq 10\tau + \tau^2 \text{ metres}$$

after τ seconds, which exceeds 5 m after about 0.45 s, well inside the 1.024 s window.

This level was run with `TRACKER_GLITCHES 1`.

Table 7: Level 4 Hardware Results with Glitches Enabled

Metric	Official Measured	Official Truth	Practice B Measured	Practice B Truth
GLITCHES	739	739 (exact)	727	727 (exact)
G	9.810 m/s ²	9.810	3.711 m/s ²	3.710
APOGEE	8.002 s, 626.08 m	8.002 s, 626.08 m	11.321 s, 1737.75 m	11.321 s, 1737.74 m
DEPLOY	12.000 s, 547.68 m	12.000 s, 547.68 m	23.999 s, 1439.54 m	24.000 s, 1439.52 m
DESCENT	-4.500 m/s	-4.500	-8.500 m/s	-8.500
LAND	133.706 s	133.707 s	193.351 s	193.355 s
Worst reading	6416 cycles	≤ 12,000	6501 cycles	≤ 12,000
State memory	2336 B	≤ 4096 B	2336 B	≤ 4096 B



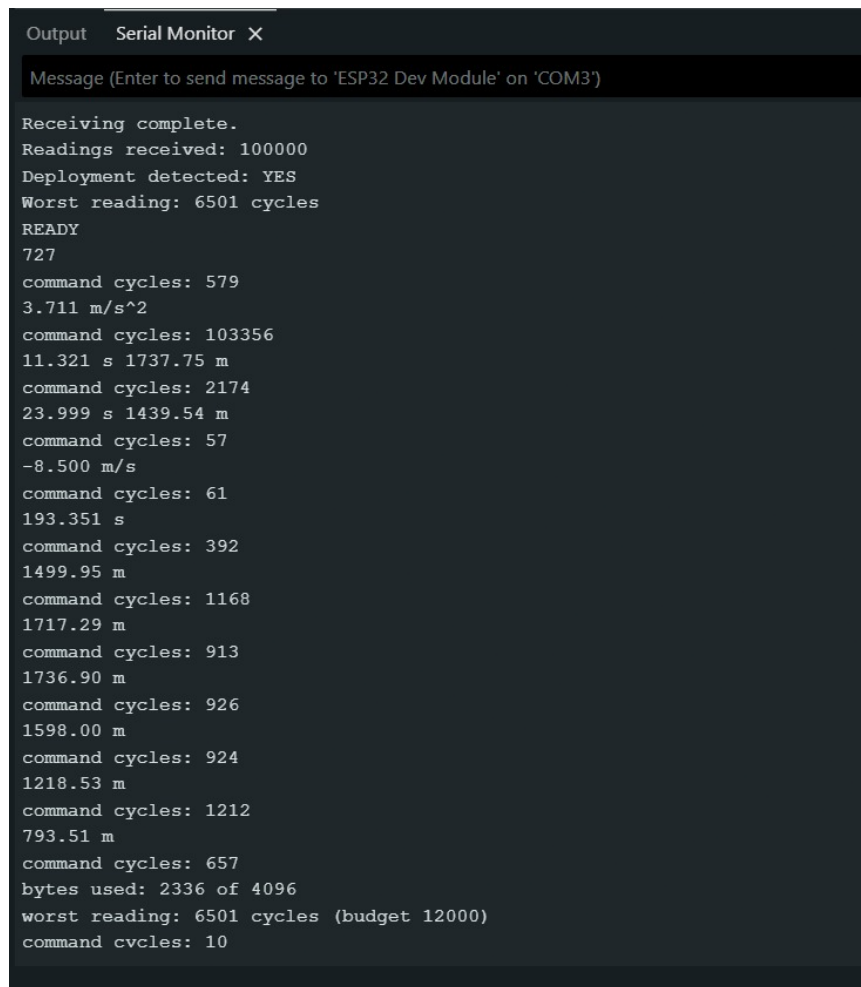
```

Output Serial Monitor X
Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')

Receiving complete.
Readings received: 100000
Deployment detected: YES
Worst reading: 6416 cycles
READY
739
command cycles: 579
9.810 m/s^2
command cycles: 102440
8.002 s 626.08 m
command cycles: 2243
12.000 s 547.68 m
command cycles: 57
-4.500 m/s
command cycles: 61
133.706 s
command cycles: 391
311.99 m
command cycles: 2058
626.08 m
command cycles: 937
547.68 m
command cycles: 1198
511.68 m
command cycles: 632
376.68 m
command cycles: 647
151.68 m
command cycles: 678
bytes used: 2336 of 4096
worst reading: 6416 cycles (budget 12000)
command cycles: 10

```

Figure 8: Level 4, Official Tracker (glitches ON): 739 glitches detected (exact); apogee and deployment altitude within 0.02 m of truth.



The screenshot shows a Serial Monitor window with a dark background and white text. At the top, there is a title bar with 'Output', 'Serial Monitor', and a close button 'X'. Below the title bar is a light gray input field with the placeholder text 'Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')'. The main area of the window displays the following text:

```
Receiving complete.  
Readings received: 100000  
Deployment detected: YES  
Worst reading: 6501 cycles  
READY  
727  
command cycles: 579  
3.711 m/s^2  
command cycles: 103356  
11.321 s 1737.75 m  
command cycles: 2174  
23.999 s 1439.54 m  
command cycles: 57  
-8.500 m/s  
command cycles: 61  
193.351 s  
command cycles: 392  
1499.95 m  
command cycles: 1168  
1717.29 m  
command cycles: 913  
1736.90 m  
command cycles: 926  
1598.00 m  
command cycles: 924  
1218.53 m  
command cycles: 1212  
793.51 m  
command cycles: 657  
bytes used: 2336 of 4096  
worst reading: 6501 cycles (budget 12000)  
command cycles: 10
```

Figure 9: Level 4, Practice Tracker B (glitches ON): 727 glitches detected (exact); apogee and deployment altitude within 0.02 m of truth.

8 Level 5: Optimization, Latency Profiling and Code Freeze

8.1 Final Optimizations

- **Closed-form Faulhaber power sums:** Loops that built $\sum k^2, \sum k^3, \sum k^4$ were replaced by exact constant-time algebraic identities (`prefixPower`), removing cycle jitter and loop cost.
- **Single-pass curve caching:** The full 3×3 normal-matrix solution runs once, in `finalizeDeployment()`, after ingestion finishes; later commands evaluate cached coefficients. Fast lookups such as `DEPLOY` (57 cycles), `DESCENT` (61), `GLITCHES` (13) and `STATS` (10) take under 100 cycles; polynomial evaluations for `LAND` (391), `ALT t` (632–1212) and `APOGEE` (2175–2244) take hundreds to a few thousand cycles (about 2–9 μs).
- **Timing headroom:** The worst reading took 6418 cycles on the Official Tracker and 6502 cycles on Practice Tracker B, leaving 45.8% (5498 cycles) free per reading.

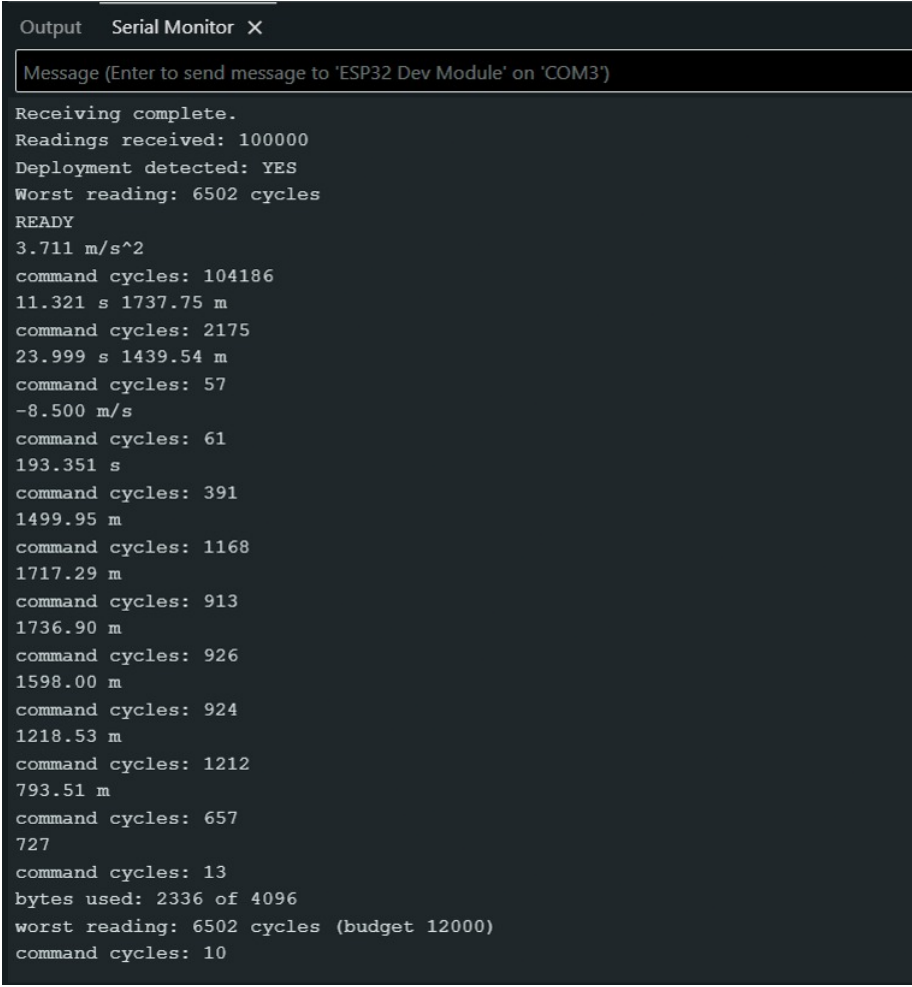
Table 8: Level 5 Final Benchmark (Physical ESP32, 240 MHz); Latencies in Cycles ($\mu\text{s} = \text{cycles} \div 240$)

Command / Metric	Official	Cycles	Practice B	Cycles
G (first call)	9.810 m/s ²	104,337	3.711 m/s ²	104,186
APOGEE	8.002 s, 626.08 m	2244	11.321 s, 1737.75 m	2175
DEPLOY	12.000 s, 547.68 m	57	23.999 s, 1439.54 m	57
DESCENT	-4.500 m/s	61	-8.500 m/s	61
LAND	133.706 s	391	193.351 s	391
ALT 0 s	311.99 m	1212	1499.95 m	1168
ALT 8 s	626.08 m	937	1717.29 m	913
ALT 12 s	547.68 m	1200	1736.90 m	926
ALT 20 s	511.68 m	632	1598.00 m	924
ALT 50 s	376.68 m	647	1218.53 m	1212
ALT 100 s	151.68 m	678	793.51 m	657
GLITCHES	739	13	727	13
STATS memory	2336 B / 4096 B	10	2336 B / 4096 B	10
STATS worst reading	6418 cycles	–	6502 cycles	–

The first `G` command costs about 104,000 cycles because it performs the final matrix fit after all readings have arrived. The 12,000-cycle limit applies to ingesting a reading during the flight, so this post-flight cost does not violate it.

```
2077 77 Normal Character .  
Output Serial Monitor X  
Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')  
Receiving complete.  
Readings received: 100000  
Deployment detected: YES  
Worst reading: 6418 cycles  
READY  
9.810 m/s^2  
command cycles: 104337  
8.002 s 626.08 m  
command cycles: 2244  
12.000 s 547.68 m  
command cycles: 57  
-4.500 m/s  
command cycles: 61  
133.706 s  
command cycles: 391  
311.99 m  
command cycles: 1212  
626.08 m  
command cycles: 937  
547.68 m  
command cycles: 1200  
511.68 m  
command cycles: 632  
376.68 m  
command cycles: 647  
151.68 m  
command cycles: 678  
739  
command cycles: 13  
bytes used: 2336 of 4096  
worst reading: 6418 cycles (budget 12000)  
command cycles: 10
```

Figure 10: Level 5, Official Tracker: final firmware answering all commands; worst reading 6418 cycles, 2336 bytes.



```
Output Serial Monitor X
Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')

Receiving complete.
Readings received: 100000
Deployment detected: YES
Worst reading: 6502 cycles
READY
3.711 m/s^2
command cycles: 104186
11.321 s 1737.75 m
command cycles: 2175
23.999 s 1439.54 m
command cycles: 57
-8.500 m/s
command cycles: 61
193.351 s
command cycles: 391
1499.95 m
command cycles: 1168
1717.29 m
command cycles: 913
1736.90 m
command cycles: 926
1598.00 m
command cycles: 924
1218.53 m
command cycles: 1212
793.51 m
command cycles: 657
727
command cycles: 13
bytes used: 2336 of 4096
worst reading: 6502 cycles (budget 12000)
command cycles: 10
```

Figure 11: Level 5, Practice Tracker B: final firmware answering all commands; worst reading 6502 cycles, 2336 bytes.

9 State Struct Memory Architecture (2336 B of 4096 B)

All persistent state lives in one `struct GroundState state`. The compiler reports a total of 2336 bytes (1760 bytes free, 57.0% used). The declared fields add up to 2317 bytes; the remaining 19 bytes are compiler alignment padding (64-bit integers and doubles are 8-byte aligned). A compile-time assertion `static_assert(sizeof(GroundState) <= 4096)` strictly enforces the limit.

Table 9: Field-by-Field Breakdown of GroundState

Field(s)	Type	Bytes	Purpose and Justification
<code>readingsReceived</code>	<code>uint32_t</code>	4	reading counter (to 100,000)
<code>worstCycles</code>	<code>uint32_t</code>	4	peak cycles via <code>ESP.getCycleCount()</code>
<code>firstN</code>	<code>uint32_t</code>	4	count for the first 8-second fit
<code>firstSumQ2, firstSumY</code>	<code>int64_t × 2</code>	16	64-bit sums $\sum q^2, \sum y$ (no overflow)
<code>firstSumQY, firstSumQ2Y</code>	<code>int64_t × 2</code>	16	64-bit cross sums for the coast fit
<code>firstA, firstB, firstC</code>	<code>double × 3</code>	24	coast-parabola coefficients (also used by the trigger)
<code>firstReady</code>	<code>bool</code>	1	8-second fit complete
<code>coastN, coastLastIndex</code>	<code>uint32_t × 2</code>	8	count and end index of the coast fit
<code>coastSumY, XY, X2Y</code>	<code>int64_t × 3</code>	24	normal sums of the refined coast fit
<code>paraN</code>	<code>uint32_t</code>	4	count for the descent line
<code>paraSumY, X, X2, XY</code>	<code>int64_t × 4</code>	32	normal sums of the descent line
<code>ring[1024]</code>	<code>uint16_t × 1024</code>	2048	1024 clean readings in 5 cm units
<code>rawHistory[5]</code>	<code>uint16_t × 5</code>	10	window of the causal median
<code>rawHistoryCount, Pos</code>	<code>uint8_t × 2</code>	2	median-window pointers
<code>glitchCount</code>	<code>uint32_t</code>	4	exact glitch counter
<code>deploymentDetected, Finalized</code>	<code>bool × 2</code>	2	deployment state flags
<code>detectionIndex, frozenStart</code>	<code>uint32_t × 2</code>	8	indices bounding the transition window
<code>deploymentTime, deploymentAltitude</code>	<code>double × 2</code>	16	evaluated deployment coordinates
<code>coastA, B, C, Center</code>	<code>double × 4</code>	32	cached coast parameters
<code>paraA, B, Center</code>	<code>double × 3</code>	24	cached descent-line parameters
<code>commandBuffer[32]</code>	<code>char × 32</code>	32	non-blocking serial input buffer
<code>commandLength, commandOverflow</code>	<code>uint8_t, bool</code>	2	command length and overflow flag
Sum of declared fields		2317	
Compiler alignment padding (2336 – 2317)		19	8-byte alignment of <code>int64_t</code> / <code>double</code>
Total reported by STATS		2336	1760 B free, 57.0% used

9.1 Grouped Memory Map of GroundState

Table 10: Grouped Memory Map (Sums to 2336 B)

Group	Bytes	Group	Bytes
Counters (<code>readingsReceived</code> , <code>worstCycles</code>)	8	Median window (<code>rawHistory</code> , <code>count</code> , <code>pos</code>)	12
First-fit sums and coefficients, <code>firstReady</code>	61	Glitch counter	4
Coast sums, <code>count</code> , last index	32	Deployment state and result	26
Descent sums and <code>count</code>	36	Cached coefficients (coast 4 + descent 3 doubles)	56
Ring buffer	2048	Serial parser (<code>buffer</code> , <code>length</code> , <code>overflow</code>)	34
Alignment padding (derived: $2336 - 2317$)	19		
Total	2336		

10 Resource Progression and Judging Compliance

Table 11: Resource Progression, Levels 0–5 (Physical ESP32)

Level	State RAM	RAM Free	Worst Reading (Cycles)	Budget Used
Level 0 (bring-up)	44 B	4052 B	903	7.53%
Level 1 (descent)	88 B	4008 B	992	8.27%
Level 2 (coast)	136 B	3960 B	2404 / 3343	20.03% / 27.86%
Level 3 (deployment)	2096 B	2000 B	2662	22.18%
Level 4 (glitches on)	2336 B	1760 B	6416 / 6501	53.47% / 54.18%
Level 5 (final freeze)	2336 B	1760 B	6418 / 6502	53.48% / 54.18%

Where two cycle values are shown they are Official / Practice B.

Table 12: Compliance with the Judging Limits (Final Level 5 Results)

Metric	Limit	Official Error	Practice B Error	Status
G	$\pm 0.02 \text{ m/s}^2$	0.000	0.001	PASS
APOGEE time	$\pm 0.02 \text{ s}$	0.000	0.000	PASS
APOGEE altitude	$\pm 0.25 \text{ m}$	0.00	0.01	PASS
DEPLOY time	$\pm 0.02 \text{ s}$	0.000	0.001	PASS
DEPLOY altitude	$\pm 0.25 \text{ m}$	0.00	0.02	PASS
DESCENT velocity	$\pm 0.01 \text{ m/s}$	0.000	0.000	PASS
LAND touchdown	$\pm 0.50 \text{ s}$	0.001	0.004	PASS
ALT t ($0 \leq t \leq 100$)	$\pm 0.25 \text{ m}$	≤ 0.01	≤ 0.05 (ALT 0)	PASS
GLITCHES	exact	739 / 739	727 / 727	PASS
Timing (worst reading)	$\leq 12,000 \text{ cycles}$	6418 (53.5%)	6502 (54.2%)	PASS
Memory	$\leq 4096 \text{ B}$	2336 B (57.0%)	2336 B (57.0%)	PASS

11 Answers to the FCA Guided Questions (Q1–Q15)

Q1 (Level 0: theoretical baseline). *How will you know an answer is right when you cannot keep the readings? Work out the true answers by hand.* From the Official Tracker secrets ($H_0 = 312$ m, $V_0 = 78.5$ m/s, $g = 9.81$ m/s², $t_D = 12$ s, $v_D = -4.5$ m/s): $t_{\text{apogee}} = V_0/g = 8.002$ s; $H_{\text{apogee}} = H_0 + V_0^2/(2g) = 626.08$ m; deployment at 12.000 s, 547.68 m; descent -4.500 m/s; landing $12.0 + 547.68/4.50 = 133.707$ s (theoretical reference truth). These are the benchmarks the measured outputs are compared with.

Q2 (Level 1: online line fit in 4 KB). Linear regression $y = A + Bk$ needs only five running accumulators: $N, \sum k, \sum y, \sum k^2, \sum ky$. Each reading updates $\sum y$ and $\sum ky$ in $O(1)$; because the k values are fixed, $\sum k$ and $\sum k^2$ are known in advance. No raw readings are stored.

Q3 (Level 1: accumulator type and overflow). For $N = 55,000$ and $k \in [-27500, +27500]$, $\sum k^2 = 1.386 \times 10^{13}$. A 32-bit signed integer overflows at 2.14×10^9 ; a float has 24 mantissa bits (exact integers only to 1.68×10^7) and truncates; a signed 64-bit integer holds up to 9.22×10^{18} and stores 1.386×10^{13} exactly. We use `int64_t`.

Q4 (milliseconds vs seconds). Yes. In integer milliseconds k is exact, so `int64` arithmetic is exact and deterministic. In seconds, $t = i/1000.0$ introduces binary floating-point fractions ($1/1000$ has no exact binary form) and accumulates rounding drift over 100,000 additions.

Q5 (Level 2: parabolic least squares). Minimising $\sum (y_i - a - bq_i - cq_i^2)^2$ gives a 3×3 system. With $q = 2i - 7999$, $\sum q = \sum q^3 = 0$, so the system splits into one equation for b and a 2×2 block for (a, c) , solved in closed form with no matrix inversion.

Q6 (physics from coefficients). With $q = 2000t - 7999$, $d^2y/dt^2 = 4 \times 10^6 c$ m/s², so $g = -80000c$ m/s². Setting $dy/dq = b + 2cq = 0$ gives $q_{\text{apogee}} = -b/(2c)$, $t_{\text{apogee}} = (7999 + q_{\text{apogee}})/2000$ s and $h_{\text{apogee}} = a + bq_{\text{apogee}} + cq_{\text{apogee}}^2$.

Q7 (centred vs time from zero). Starting at 0, all power sums up to $\sum i^4 \approx 6.55 \times 10^{18}$ are coupled in a full 3×3 system and the elimination subtracts nearly equal numbers; in 32-bit float this cancels and gave nonsense gravity ($g \approx -45.2$ and $+112$ m/s² in our first attempt, Section 12.1). Centring makes the odd sums vanish, which decouples the system, and with `int64` sums plus double division the measured gravity was 9.814 m/s² against 9.810 (Official) and 3.709 against 3.710 (Practice B) at Level 2, and $9.810 / 3.711$ at Level 5.

Q8 (Level 3: real-time deployment detection). At deployment the rocket falls at least 10 m/s faster than its terminal parachute speed (G2). The ballistic prediction keeps accelerating downward while the real altitude follows the parachute, so the altitude change over 1024 readings (1.024 s of simulated flight) lags the prediction. In the tracker model the lag is at least $10\tau + \tau^2$ m after τ seconds, which exceeds the 5 m threshold after about 0.45 s, and the threshold also exceeds the worst-case 400 cm noise difference between two readings (derived reasoning; the trigger fired correctly on both trackers).

Q9 (removing contaminated readings). We avoid retrospective subtraction. The coast accumulators are fed with a 1024-reading delay from the clean ring buffer (they only ingest reading $i - 1024$). When deployment is confirmed they are simply frozen, so no transitional or parachute-contaminated reading ever entered the coast fit, at zero subtraction cost.

Q10 (choosing the physical intersection). Equating the coast parabola and descent line gives a downward-opening quadratic ($c < 0$) with two roots. The earlier root is the descent line extrapolated backwards through the ascending rocket and is unphysical; the later root, after apogee (G2), is the true deployment.

Q11 (effect of unfiltered glitches). One 300 m (30,000 cm) glitch in $N = 10,000$ readings shifts a mean-type estimate by $30,000/10,000 = 3$ cm. In the worst case 700 glitches of 300 m with the same sign in 100,000 readings shift the mean by $700 \times 30,000/100,000 = 210$ cm = 2.1 m, and, since the descent velocity limit is only ± 0.01 m/s, even partial bias matters. Glitch rejection is therefore mandatory.

Q12 (glitch discrimination). By G3/G4, normal noise is within ± 2 m, consecutive readings differ by at most 4 m, and glitches are at least 50 m and at least 10 readings apart. A 5-point median isolates single spikes. A glitch lasts one reading, whereas deployment is a persistent shift lasting for the rest of the flight (88,000 readings on the Official Tracker).

Q13 (worst-case reading). The slowest reading occurs when ring-buffer management, the median insertion sort and the deployment threshold test coincide. Evaluating polynomial power sums with loops would add large cycle spikes; closed-form Faulhaber identities, and deferring the matrix solution until ingestion ends, bound the worst reading at 6502 cycles (54.2% of 12,000) on the hardware.

Q14 (float vs double on the ESP32). The Xtensa LX6 has a single-precision hardware FPU (a float add takes about 1 cycle); double operations are emulated in software and take roughly 35–60 cycles. Double is used only where precision is essential: once inside the streaming loop at reading 8000 to compute the initial coast fit (`calculateFirstFit()`), and in the post-flight matrix solution. The per-reading path is integer- and float-based.

Q15 (the nastiest flight). Analysis only; we did not run a synthetic extreme-flight test. The most demanding flight allowed by G1–G4 combines minimal gravity ($g = 2 \text{ m/s}^2$) with high launch velocity (altitudes near the 3000 m ceiling, where the 5 cm quantization and the 3000 m clamp matter most), a late parachute at $t_D = 45 \text{ s}$ with the minimum velocity step $\Delta v = 10 \text{ m/s}$ (the smallest divergence signal) and glitches at the maximum density (1% of readings, 10 readings apart). Our design addresses these by design: the median strips spikes before the ring buffer, the 500 cm threshold exceeds the 400 cm worst-case noise difference, and int64 accumulators are exact to 9.22×10^{18} . These properties are argued, not demonstrated on hardware.

12 Engineering Challenges, Failures and Solutions

12.1 Catastrophic Floating-Point Cancellation in Level 2

Failure. The first Level 2 coast fit returned wildly varying gravity ($g \approx -45.2 \text{ m/s}^2$ or $+112.0 \text{ m/s}^2$) on the Official Tracker. **Root cause.** The denominator $N \cdot \sum q^4 - (\sum q^2)^2$ (about 2.3×10^{22} , a difference of numbers of order 10^{22} , with $\sum q^4 \approx 6.55 \times 10^{18}$) was evaluated in single-precision float, whose 24-bit mantissa cannot represent it. **Solution (Gana).** All power accumulators became 64-bit integers (`int64_t`) and the division was done in 64-bit double; gravity stabilised to 9.814 m/s^2 (within 0.004 m/s^2 of truth).

12.2 RAM Overflow in the Initial Ring-Buffer Design

Failure. An early Level 4 design used a ring of 2048 raw 32-bit floats (8192 bytes), pushing `sizeof(GroundState)` to 8640 bytes, double the 4096-byte limit. **Solution (Samudra and Sindhu).** Since readings are non-negative and bounded by 3000 m (G1), altitude was quantized to 5 cm ($v = (h_{\text{cm}} + 2)/5$) so it fits a `uint16_t`, and the ring was cut to 1024 elements: 2048 bytes, bringing the total to 2336 bytes (1760 free).

12.3 Boundary Distortion Around Parachute Inflation

Failure. Fitting curves over the whole flight without masking the opening distorted the descent slope ($v_D \approx -5.1 \text{ m/s}$ instead of -4.5 m/s). **Root cause.** During inflation (500–1000 ms) the rocket undergoes transient deceleration that fits neither ballistic flight nor steady descent. **Solution.** A 1024-reading exclusion window around the detected opening keeps transitional readings out of both accumulator sets.

12.4 Earlier Level 3 Cost of the G Command

The Level 3 G command took 1,247,112 cycles (Official) and 2,118,771 cycles (Practice B). Closed-form power sums and curve caching in Level 5 reduced it to 104,337 and 104,186 cycles (post-flight only).

13 LLM Usage Audit Log and Key Learnings

Table 13: Audit Log of LLM Queries, Suggestions and Physical Verification

Prompt / Task	LLM Suggestion	How the Team Verified	Outcome
Closed-form power-sum identities for $\sum k^2, \sum k^3, \sum k^4$	Faulhaber polynomial formulas	Verified by Gana on paper and benchmarked in desktop C++	Used in the final code
5-element median filter	9-comparator branchless sorting network	Tested by Sindhu against 100,000 random vectors	Implemented as a clean 5-value insertion sort
Hallucination / failure: detecting deployment via local variance	Claimed deployment could be detected by a sudden spike in rolling variance	Run on the board	Failed completely: normal noise ($\pm 2 \text{ m}$) caused continuous false triggers. Discarded by Gana in favour of trajectory-divergence lag
L ^A T _E X table and layout formatting	<code>booktabs</code> formatting, clean column alignment	Compiled and visually audited by Brunda	Used in the report

13.2 Key Engineering Learnings

- **Memory and alignment:** A float ring buffer would have used the entire 4096 B; 5 cm `uint16_t` halves it. In a struct on a 32-bit Xtensa target, 64-bit integers and doubles are 8-byte aligned, so field order matters: our final struct has 19 bytes of padding on top of 2317 bytes of fields.
- **Hardware FPU vs software emulation:** The LX6 handles 32-bit floats in hardware but emulates doubles in software. Except for the one-time 8-second fit at reading 8000 and the post-flight regression, the streaming loop avoids double operations.
- **Coordinate centring:** Centring time makes odd power sums vanish and decouples the normal equations; with `int64` sums and double division it made a high-precision fit possible on a microcontroller.
- **Measure, don't guess:** Cycle counts came from `ESP.getCycleCount()` on every run, and every result was compared with tracker truth.

14 Limitations, Conclusion and Appendix

14.1 Known Limitations

- Altitude values are clamped at 60,000 units of 5 cm (3000 m), which matches G1 but leaves no margin above it.
- The glitch test is active only from reading 1000 (glitches start at $t \geq 1.0$ s).
- The first G command costs about 104,000 cycles; it runs after the flight and is outside the per-reading deadline.
- Results were verified on two trackers (Official and Practice B). No unseen-tracker or synthetic extreme-flight run is included in this report.
- The 500 cm deployment margin over the worst-case 400 cm noise difference is small (1 m); it was sufficient on both trackers but is the tightest margin in the design.

14.2 Conclusion

Team ByteForce designed, implemented and verified on physical hardware a complete autonomous ground station for the FCA “Catch the Rocket” hackathon. Without hard-coded flight constants, the firmware extracted gravity, apogee, deployment, descent rate, landing time and trajectory altitudes on both the Official Tracker and Practice Tracker B, within a small fraction of the judging tolerances, using 2336 bytes of state (57.0% of 4096) and at most 6502 cycles per reading (54.2% of 12,000).

14.3 Appendix: Verified Code and Artefacts in Team-ByteForce.zip

- `LEVEL_0_FINAL.ino`: bring-up skeleton, cycle profiler, non-blocking serial parser.
- `LEVEL_1_OFFICIAL.ino`, `LEVEL_1_PRACTICE.ino`: centred linear descent regression.
- `LEVEL_2_OFFICIAL.ino`, `level2_practice.ino`: decoupled ballistic coast fit.
- `LEVEL_3_OFFICIAL.ino`, `LEVEL_3_PRACTICE.ino`: 100-block moment compression and deployment root solver.
- `LEVEL_4_OFFICIAL.ino`, `LEVEL_4_PRACTICE.ino`: causal 5-point median filter, ring buffer, real-time trigger.
- `LEVEL_5_OFFICIAL.ino`, `LEVEL_5_PRACTICE.ino`: final firmware with Faulhaber power sums and curve caching.
- `outputs.docx`: the 11 serial-monitor captures (Figures 1–11) and the team roster.
- `fca_hackathon_catch_the_rocket.pdf`: the problem specification and judging criteria.

Chaithra Gana G	Sindhu GV	Samudra Kar	Brunda
Skeptic and Code Integrate	Builder	Measurer	Scribe